3 **Solution:** By Defn, $x = 3\begin{bmatrix} 1 \\ .2 \end{bmatrix} + 2\begin{bmatrix} .2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.4 \\ 2.6 \end{bmatrix}$

3. Could R^3 possibly contain a four-dimensional subspace? Explain.

Solution: A four-dimensional subspace by defn. would contain a basis of four linearly independent vectors.

. This is impossible inside R^3.

. Since any linearly independent set in R^3 has no more than three vectors, any subspace of R^3 has dimension no more than 3. □